Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 100307 *Subject: Electrical Machine-I* *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Why armature of a dc machine is laminated?
(i) To reduce hysteresis loss.
(ii) To reduce eddy current loss.
(iii) To reduce hysteresis loss and eddy current loss.
(iv) None of the above.

(b) Commutator in DC generator is used for
(i) collecting of current (ii) reduce losses
(iii) increase efficiency (iv) convert AC armature current in to DC

(c) For lap winding, no of parallel path is
(i) 2 (ii) no of poles
(iii) 4 (iv) None

(d) Inter pole winding is connected in
(i) series with armature (ii) series with main poles
(iii) parallel with armature (iv) parallel with main poles Inter

(e) Dummy coils are used for
(i) increasing efficiency (ii) reducing armature reaction
(iii) mechanical balancing (iv) all of the above

(f) Which of the following motors has negative speed regulation?
(i) Shunt motors. (ii) Series motors.
(iii) Differential compound motors. (iv) Both B and C.

(g) Which of the following statements is/are correct statements?
(i) EMF per turn in HV winding is more than EMF per turn in LV winding
(ii) EMF per turn in HV winding is less than EMF per turn in LV winding
(iii) EMF per turn in both the windings are equal
(iv) all of the above

(h) Tick the correct alternative MMF =
(i) Flux × Reluctance (ii) Flux / Reluctance
(iii) Flux + Reluctance (iv) Flux - Reluctance

(i) In a transformer zero voltage regulation is achieved at a load power factor which is
(i) zero (ii) unity (iii) lagging (iv) leading

Q.2 (a) Write and explain parallel operation of single–phase transformer with its necessity. **[7]**

(b) List out the merits and demerits of core and shell type transformer. **[7]**

Q.3 (a) Draw and explain phasor diagram of single phase transformer and derive the expression of induced emf in transformer. **[7]**

(b) A transformer with 800 primary turns and 200 secondary turns supplied from a 100V ac. Calculate secondary voltage and voltage per turn. **[7]**

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Q.4 (a) Briefly discuss different parts of DC generator with diagram. [7]

(b) A 4 pole generator with wave wound armature has 51 slots each having 48 conductors. The flux per pole is 7.5 mWb. At what speed induced emf will be 440V. [7]

Q.5 (a) What is armature reaction in dc generator? What are the effects of armature reaction in dc generator? How to improve those effects? [8]

(b) A long shunt compound generator delivers a load current of 50 A at 500 V. Resistance of armature, series field and shunt fields are 0.5 ohm, 0.03 ohm and 250 ohm respectively. Calculate the generated emf and armature current. [6]

Q.6 (a) Draw and explain the typical magnetic circuit with air and its equivalent electric circuit. [7]

(b) Specify the causes for Hysteresis and Eddy current losses in Electrical machines. Also give the methods in construction to minimize the above losses. [7]

Q.7 Draw the torque armature current characteristics of different types of dc motor. What are the different methods of speed control of dc shunt motor? Explain them. [14]

Q.8 (a) Draw and explain the B-H curve of DC generator. [7]

(b) Explain linear and non-linear magnetic circuit. [7]

Q.9 Write short notes on ***any two*** of the following: [7x2=14]

(a) Auto transformer

(b) Losses in DC machine

(c) Open and short circuit tests of transformer

(d) Commutation in DC machine

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